H.W # 03

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(Q: 01)

$$x'(t) = 2x + 3y$$
$$y'(t) = 2x + y$$

$$\begin{pmatrix} x(t) \\ y(t) \end{pmatrix}' = \begin{bmatrix} 2 & 3 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

$$\begin{bmatrix} 2 & 3 \\ 2 & 1 \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix} = \begin{bmatrix} 2-\lambda & 3 \\ 2 & 1-\lambda \end{bmatrix}$$

$$\begin{vmatrix} 2-\lambda & 3 \\ 2 & 1-\lambda \end{vmatrix} = 0$$

$$2-\lambda(1-\lambda) - 6 = 0$$
$$2 - 2\lambda - \lambda + \lambda^2 - 6 = 0$$
$$\lambda^2 - 3\lambda - 4 = 0$$
$$\lambda = \frac{3 \pm \sqrt{9+16}}{2}$$
$$= \frac{3 \pm \sqrt{25}}{2}$$
$$\lambda = 4 \text{ or } \lambda = -1$$

for $\lambda = 4$

$$\left[\begin{array}{cc|c} -2 & 3 & 0 \\ 2 & -3 & 0 \end{array}\right] \sim \left[\begin{array}{cc|c} -2 & 3 & 0 \\ 0 & 0 & 0 \end{array}\right]$$

$$x_1 = \frac{3}{2} y$$
$$y = y$$
$$x = \begin{bmatrix} 3/2 \\ 1 \end{bmatrix} y$$

for $\lambda = -1$

$$\left[\begin{array}{cc|c} 3 & 3 & 0 \\ 2 & 2 & 0 \end{array}\right] \sim \left[\begin{array}{cc|c} 1 & 1 & 0 \\ 0 & 0 & 0 \end{array}\right]$$

$$x_1 = -y$$
$$y = y$$
$$x = \begin{bmatrix} -1 \\ 1 \end{bmatrix} y$$

$$x(t) = c_1 e^{4t}\begin{bmatrix} 3/2 \\ 1 \end{bmatrix} + c_2 e^{-t}\begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

(Q:2)

$$X' = \begin{bmatrix} 1 & -2 & 2 \\ -2 & 1 & -2 \\ 2 & -2 & 1 \end{bmatrix} X$$

$$\begin{vmatrix} 1-\lambda & -2 & 2 \\ -2 & 1-\lambda & -2 \\ 2 & -2 & 1-\lambda \end{vmatrix} = 0$$

$$1-\lambda\left((1-\lambda)^2-4\right)+2\left(-2(1-\lambda)+4\right)+2\left(4-2(1-\lambda)\right)=0$$

$$1-\lambda\left(1+\lambda^2-2\lambda-4\right)+2\left(-2+2\lambda+4\right)$$

$$+2\left(4-2+2\lambda\right)=0$$

$$\cancel{1}+\cancel{\lambda^2}-\cancel{2\lambda}-\cancel{4}-\cancel{\lambda}-\cancel{\lambda^3}+\cancel{2\lambda^2}+\cancel{4\lambda}-\cancel{4}+\cancel{4\lambda}+\cancel{8}$$

$$+\cancel{8}-\cancel{4}+\cancel{4\lambda}=0$$

$$-\lambda^3+3\lambda^2+9\lambda+5=0$$

$$\lambda^3-3\lambda^2-9\lambda-5=0$$

$$\lambda=5 \text{ or } \lambda=-1$$

$$\begin{array}{r|rrrr} -1 & 1 & -3 & -9 & -5 \\ & & -1 & 4 & 5 \\ \hline & 1 & -4 & -5 & \big| \; 0 \end{array}$$

for $\lambda=5$

$$\begin{bmatrix} -4 & -2 & 2 \\ -2 & -4 & -2 \\ 2 & -2 & -4 \end{bmatrix} \sim \left[\begin{array}{ccc|c} +2 & +1 & -1 & 0 \\ 0 & -3 & -3 & 0 \\ 0 & -3 & -3 & 0 \end{array}\right] \sim \left[\begin{array}{ccc|c} 2 & 1 & -1 & 0 \\ 0 & -3 & -3 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

$$\left[\begin{array}{ccc|c} 1 & 1/2 & -1/2 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right] \sim \left[\quad\right]$$

$x_1 = x_3/2 + x_3/2$

$x_2 = -x_3$

$x_3 = x_3$

$$x = \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} x_3$$

for $\lambda = -1$

$$\begin{bmatrix} 2 & -2 & 2 \\ -2 & 2 & -2 \\ 2 & -2 & 2 \end{bmatrix} \sim \left[\begin{array}{ccc|c} 1 & -1 & 1 & 0 \\ -2 & 2 & -2 & 0 \\ 2 & -2 & 2 & 0 \end{array}\right] \sim \left[\begin{array}{ccc|c} 1 & -1 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

$x_1 = x_2 - x_3$

$x_2 = x_2$

$x_3 = x_3$

$$\Rightarrow \quad x = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} x_2 + \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} x_3$$

$$x(t) = c_1 e^{5t} \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} + c_2 e^{-t} \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} + c_3 e^{-t} \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$$

(Q:3)

$$X' = \begin{bmatrix} 2 & 1 & 6 \\ 0 & 2 & 5 \\ 0 & 0 & 2 \end{bmatrix} X$$

$\lambda = 2$ since triangular matrix

$$\left[\begin{array}{ccc|c} 0 & 1 & 6 & 0 \\ 0 & 0 & 5 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right] \sim \left[\begin{array}{ccc|c} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

$x_2 = 0$

$x_3 = 0 \Rightarrow x = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} x_1$

$x_1 = x_1$

$$x(t) = e^{2t} \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

(Q:4)

$x'(t) = 6x - y$
$y'(t) = 5x + 4y$

$$\begin{pmatrix} x(t) \\ y(t) \end{pmatrix}' = \begin{bmatrix} 6 & -1 \\ 5 & 4 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

$$\begin{vmatrix} 6-\lambda & -1 \\ 5 & 4-\lambda \end{vmatrix} = 0$$

$(6-\lambda)(4-\lambda) + 5 = 0$
$24 - 6\lambda - 4\lambda + \lambda^2 + 5 = 0$
$\lambda^2 - 10\lambda + 29 = 0$
$\lambda = 5+2i \qquad \lambda = 5-2i$

for $\lambda = 5+2i$

$$\left[\begin{array}{cc|c} 1-2i & -1 & 0 \\ 5 & -1-2i & 0 \end{array}\right] \Rightarrow \begin{aligned} 5x_1 &= 1+2i\,x_2 \\ x_1 &= \frac{1}{5} + \frac{2}{5} i\, x_2 \end{aligned}$$

$(1-2i)x_1 = x_2$
$(1-2i)\left(\frac{1+2i}{5}\right)x_2 - x_2 = 0$

$\frac{1^2 - (2i)^2}{5} x_2 - x_2 = 0$

$x_2 - x_2 = 0$
$x_2 = x_2$ free var.
$x_1 = \left(\frac{1+2i}{5}\right)x_2$

$$x = \begin{bmatrix} (1+2i)/5 \\ 1 \end{bmatrix} x_2$$

By theorem

$$\bar{\lambda} = \overline{5+2i}$$
$$\lambda = 5-2i \Rightarrow \bar{x} = \overline{\begin{bmatrix} (1+2i)/5 \\ 1 \end{bmatrix}} x_2$$

$$\bar{x} = \begin{bmatrix} (1-2i)/5 \\ 1 \end{bmatrix} x_2$$

$$x(t) = c_1 e^{(5+2i)t} \begin{bmatrix} (1+2i)/5 \\ 1 \end{bmatrix} + c_2 e^{(5-2i)t} \begin{bmatrix} (1-2i)/5 \\ 1 \end{bmatrix}$$

(Q:5)

$$x' = \begin{bmatrix} 2 & 8 \\ -1 & -2 \end{bmatrix} x \qquad x(0) = \begin{bmatrix} 2 \\ -1 \end{bmatrix}$$

$$\begin{vmatrix} 2-\lambda & 8 \\ -1 & -2-\lambda \end{vmatrix} = 0$$

$$-(2-\lambda)(2+\lambda) + 8 = 0$$
$$-4+\lambda^2+8 = 0$$
$$\lambda^2+4 = 0$$
$$\lambda^2 = -4$$
$$\lambda = 2i \quad \lambda = -2i$$

for $\lambda = 2i$

$$\begin{bmatrix} 2-2i & 8 \\ -1 & -2-2i \end{bmatrix}$$

$$+x_1 = (-2-2i)x_2$$
$$(2-2i)x_1 = -8x_2$$

$$(2-2i)(-2-2i)x_2+8x_2=0$$
$$-(2-2i)(2+2i)x_2+8x_2=0$$
$$-[4+4]x_2+8x_2=0$$
$$-8x_2+8x_2=0$$
$$x_2=x_2 \quad \text{// free variable}$$

$$x=\begin{bmatrix}-2-2i\\1\end{bmatrix}x_2$$

for $\lambda=-2i$

$$\bar{x}=\begin{bmatrix}-2+2i\\1\end{bmatrix}x_2$$

$$= e^{2it}\begin{bmatrix}-2-2i\\1\end{bmatrix}$$

$$= (\cos 2t + i\sin 2t)\begin{bmatrix}-2-2i\\1\end{bmatrix}$$

$$= \begin{bmatrix}-2\cos 2t - 2\cos 2t\, i - 2i\sin 2t + 2\sin 2t\\ \cos 2t + i\sin 2t\end{bmatrix}$$

$$= c_1\begin{bmatrix}-2\cos 2t + 2\sin 2t\\ \cos 2t\end{bmatrix} + \overset{c_2}{i}\begin{bmatrix}-2\cos 2t - 2\sin 2t\\ \sin 2t\end{bmatrix}$$

$$\begin{bmatrix}2\\-1\end{bmatrix} = \begin{bmatrix}-2\\1\end{bmatrix}c_1 + c_2\begin{bmatrix}-2\\0\end{bmatrix}$$

$$\begin{bmatrix}2\\-1\end{bmatrix} = \begin{bmatrix}-2 & -2\\1 & 0\end{bmatrix}\begin{bmatrix}c_1\\c_2\end{bmatrix}$$

$$c_1 = -1$$
$$-2c_1 - 2c_2 = 2$$
$$c_2 = 0$$

$$\begin{bmatrix}c_1\\c_2\end{bmatrix} = \begin{bmatrix}-1\\0\end{bmatrix}$$

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